Design and implementation of two stage CMOS operational amplifiers using 180nm technology

Rail-to-Rail CMOS OpAmp With Improved Bulk-Driven Input Stage

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Rail – to – Rail Two Stage Operational Amplifier

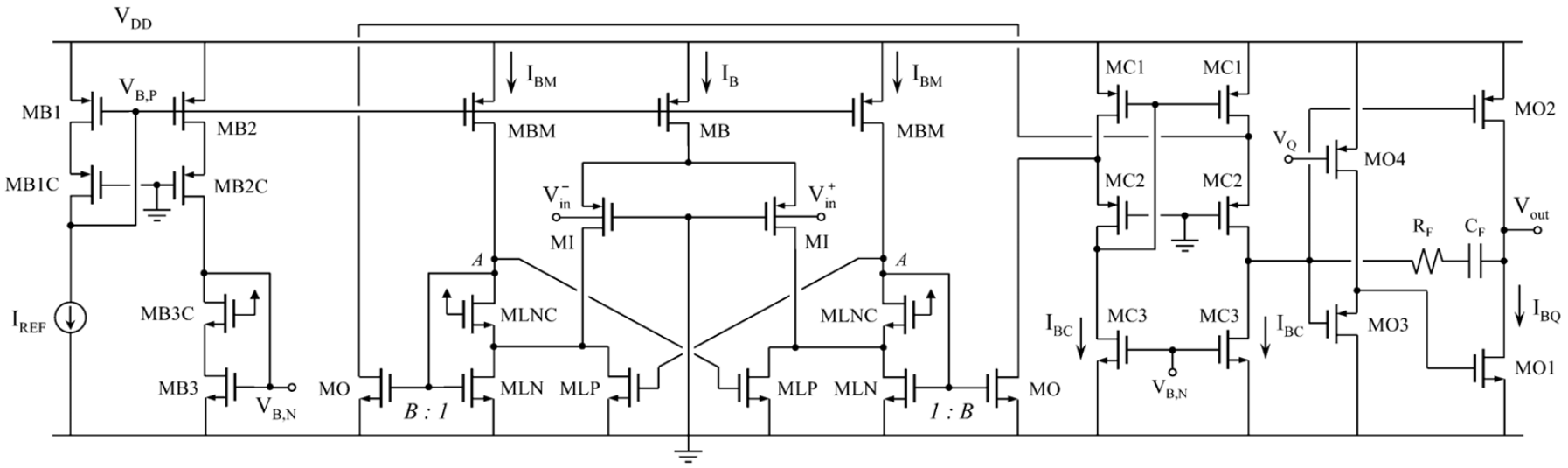
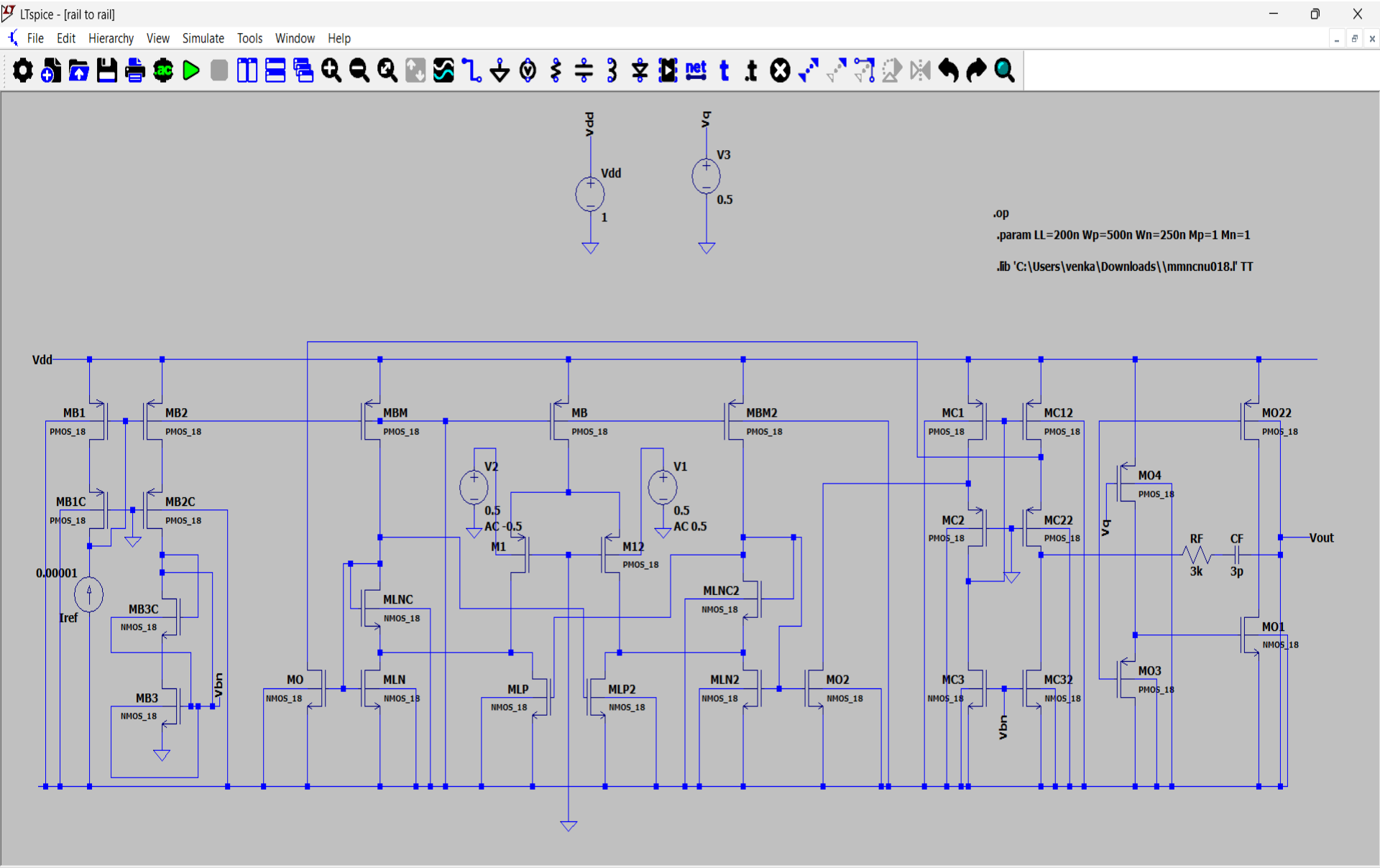
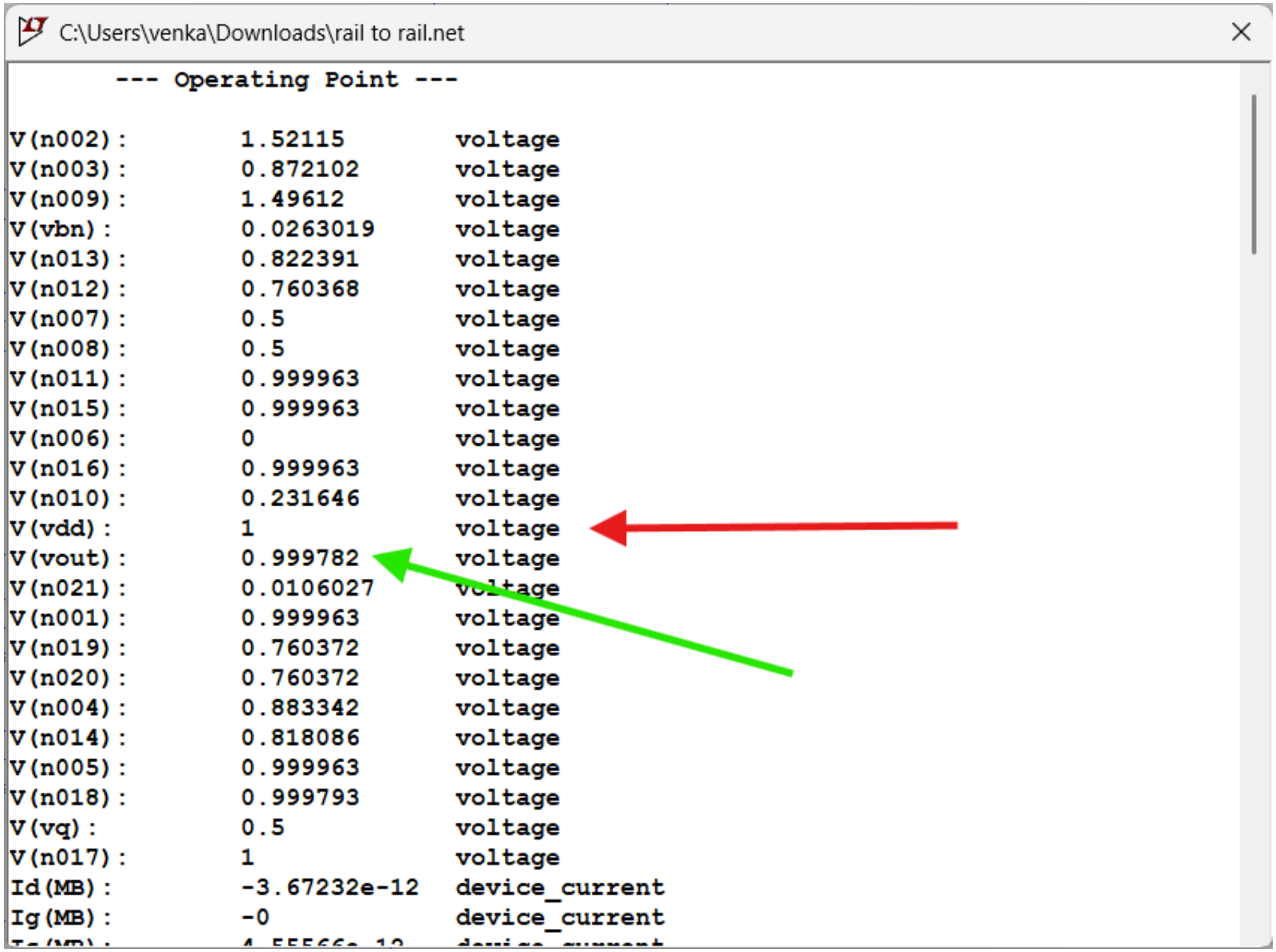


Fig. 5. 1-V rail-to-rail two-stage operational amplifier.

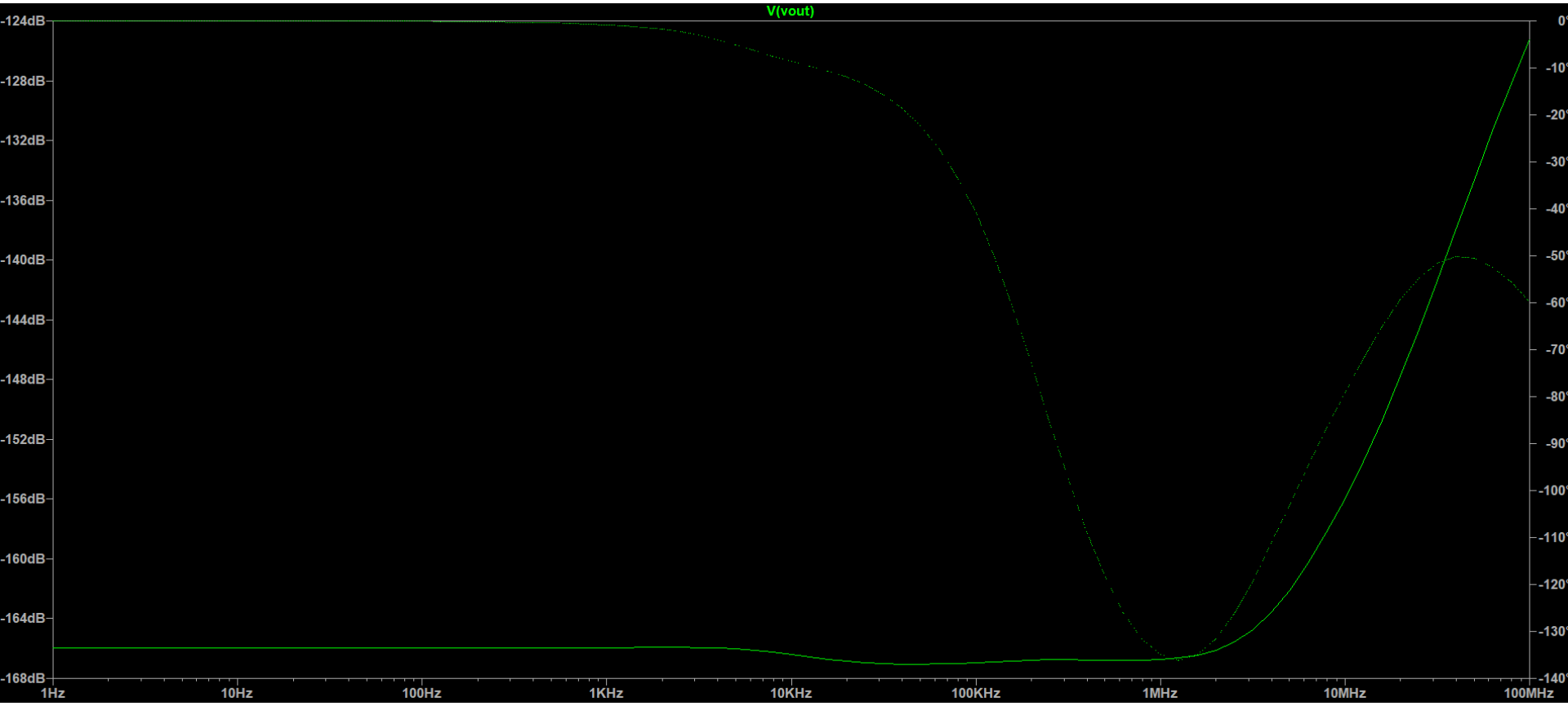
LT Spice Schematic



Operating Point



AC analysis


V(vout)
-124dB
-128dB
-132dB
-136dB
-140dB
-144dB
-148dB
-152dB
-156dB
-160dB
-164dB
-168dB
0°
-10°
-20°
-30°
-40°
-50°
-60°
-70°
-80°
-90°
-100°
-110°
-120°
-130°
-140°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz
100MHz

Thank you